Requirement Analysis

Data Flow Diagram & User Stories

Project Title	India's Agricultural Crop Production Analysis
Domain	Agriculture / Data Analytics & Business Intelligence
Tool	Tableau Desktop / Tableau Public
Dataset	Historical Indian Crop Production Data, 1997–2021
Prepared By	Divya Kauluri (Team Project — completed individually; the team lead and other assigned members did not contribute to the work)
Mentor	N/A — Individual Submission
GitHub Repository	https://github.com/divyakauluri-spec/India-s-agricultural-crop-production-analysis
Tableau Public Link	https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject
Date	02/07/2026

Project Context

This report presents a comprehensive visual exploration of India's agricultural cultivation using 25 years (1997–2021) of historical crop production data. Nine Tableau visualizations — a KPI Summary panel, State-wise Production Map, Year-wise Total Production chart, Trend of Agricultural Yield chart, Cultivated Area by Crop chart, State-wise Cultivated Area Comparison chart, Leading States by Production chart, Production Distribution by Unit Type donut, and Season-wise Production pie — are combined into four role-specific dashboards and a five-point guided Story, enabling stakeholders to uncover patterns, identify areas of growth or concern, and make data-driven decisions.

By harnessing the power of Tableau, this report presents the data in a visually appealing, interactive form that lets readers explore the intricacies of India's agricultural cultivation for themselves.

Scenario 1: Small-Scale Farmer

Rajesh is a small-scale farmer in Uttar Pradesh who wants to maximize his crop yield and income. However, he struggles to decide which crops to grow each season due to changing weather patterns and market demand. By analyzing historical agricultural data (1997–2021), Rajesh needs insights into seasonal crop trends, regional productivity, and high-performing crops to make better farming decisions and reduce risk. The Farmer's View dashboard serves Rajesh through the Cultivated Area by Crop and Season-wise Production charts.

Scenario 2: Government Policy Analyst

Ananya works with the Ministry of Agriculture and is responsible for designing policies to improve agricultural productivity across India. She needs to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations. Using Tableau dashboards, she aims to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively. The Policy Analyst's View dashboard serves Ananya through the State-wise Production Map and Trend of Agricultural Yield chart.

Scenario 3: Agribusiness Investor

Vikram is an agribusiness investor looking to invest in high-growth agricultural sectors. He needs to analyze historical crop production data to identify profitable crops, emerging trends, and regions with consistent growth. By leveraging visual analytics in Tableau, Vikram seeks to minimize investment risk and make informed decisions on where and what to invest in the agricultural market. The Investor's View dashboard serves Vikram through the Leading States by Production and Cultivated Area by Crop charts.

Data Flow Diagram

- Step 1 — Raw Source: Historical crop production data in Excel/CSV format (345,407 rows, 1997–2021)
- Step 2 — Data Cleaning & Preprocessing: Python (Pandas) used to rename columns, strip whitespace, extract numeric year from 'YYYY-YY' format, add Yield = Production/Area, remove nulls
- Step 3 — Cleaned Dataset: skillwallet_crop_cleaned.csv (345,407 rows, 12 columns) loaded into Tableau
- Step 4 — Tableau Data Source: CSV connected as a Text File source; State Name field assigned Geographic Role = State/Province
- Step 5 — Visualization Layer: nine chart sheets built (KPI panel, map, line, bar charts, donut, pie)
- Step 6 — Dashboard Layer: four role-specific dashboards assembled from chart sheets with interactive filters
- Step 7 — Story Layer: Tableau Story with five narrative points combining dashboards for a guided presentation
- Step 8 — Publishing: Workbook saved to Tableau Public; shareable URL generated for GitHub submission

User Stories

User Type	Epic	User Story	Priority	Sprint
Farmer (Rajesh)	Data Visualization	As a farmer, I want to see which crops perform best in my state during Kharif season so I can choose what to plant	High	Sprint 1
Farmer (Rajesh)	Dashboard	As a farmer, I want a single dashboard that shows me seasonal and crop trends together	High	Sprint 2
Policy Analyst (Ananya)	Data Visualization	As a policy analyst, I want to see which states have the lowest production so I can target interventions	High	Sprint 1
Policy Analyst (Ananya)	Dashboard	As a policy analyst, I want to see the map and trend chart together to compare regions and time periods	High	Sprint 2
Investor (Vikram)	Data Visualization	As an investor, I want to see which crops dominate production volume to identify investment targets	Medium	Sprint 2
Investor (Vikram)	Dashboard	As an investor, I want a combined view of crop rankings to compare opportunities	Medium	Sprint 2
All Users	Story	As a stakeholder, I want a guided narrative walkthrough of all dashboards to understand the full picture	Medium	Sprint 3
All Users	Web Integration	As a stakeholder, I want to access the dashboard from any browser without installing Tableau	High	Sprint 3